COURSE CODE: CSA1712 - ARTIFICIAL INTELLIGENCE
NAME: SHAIK MOHAMMED WASEEM RAYAN
REGD. NO: 192373004

CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

(a) Propositional Logic

Representation Symbols Used

- **F** = Patient has Fever
- **C** = Patient has Cough
- **B** = Patient has Breathlessness
- **R** = Patient has Rash
- **Flu** = Patient has Flu
- **Measles** = Patient has Measles
- **Pneumonia** = Patient has Pneumonia

Rules

1. If Fever and Cough, then Flu
   $\mathbf{F \land C \rightarrow Flu}$
2. If Fever and Rash, then Measles
   $\mathbf{F \land R \rightarrow Measles}$
3. If Cough and Breathlessness, then Pneumonia
   $\mathbf{C \land B \rightarrow Pneumonia}$

Facts for Patient A

- **F = True**
- **C = True**
- **B = True**

Therefore, the Knowledge Base is:

$\mathbf{KB = \{F, C, B, F \land C \rightarrow Flu, F \land R \rightarrow Measles, C \land B \rightarrow Pneumonia\}}$

(b) Forward

Chaining Initial Facts

{F, C, B}

Step 1

Rule fired: **F ∧ C → Flu**

Since F and C are true, derive:

Flu

Updated facts:

{F, C, B, Flu}

Step 2

Rule checked: **F ∧ R → Measles**

Rash is not available; hence no inference.

Step 3

Rule fired: **C ∧ B → Pneumonia**

Since C and B are true, derive:

Pneumonia

Updated facts:

{F, C, B, Flu, Pneumonia}

Final Diagnoses

- Flu
- Pneumonia

Measles cannot be concluded.

(c) Backward Chaining for Goal Pneumonia

Goal: **Pneumonia**

Using rule:

C ∧ B → Pneumonia

Required subgoals:

- C

- B

Both C and B are already true facts.

Goal Reduction Tree

```
          Pneumonia
              |
        -----------
        |         |
        C         B
        |         |
      True      True
```

Conclusion

Patient A has Pneumonia according to the knowledge base.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

(a) Unifica

tion Rule 1

$\forall x\ [Vehicle(x) \land EmergencyType(x) \rightarrow ClearPath(x)]$

Facts:

- Vehicle(Ambulance)
- EmergencyType(Ambulance)

Step 1

Match Vehicle(x) with Vehicle(Ambulance)

Substitution:

θ_1 = {x/Ambulance}

Step 2

Apply θ_1 to EmergencyType(x):

EmergencyType(Ambulance)

Matches the fact.

Most General Unifier

θ = {x/Ambulance}

Derived fact:

ClearPath(Ambulance)

(b) Resolution Proof for

Proceed(CarA) CNF Conversion

Rule 1:

¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

Rule 2:

¬ClearPath(x) ∨ GreenSignal(x)

Rule 3:

¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Facts:

- Vehicle(Ambulance)
- EmergencyType(Ambulance)
- Vehicle(CarA)
- Behind(CarA,Ambulance)

Negated goal:

¬Proceed(CarA)

Resolution Steps

1. From Rule 1 and facts derive **ClearPath(Ambulance)**.
2. From Rule 2 derive **GreenSignal(Ambulance)**.
3. Apply Rule 3 with x=CarA and y=Ambulance and resolve with facts to derive **Proceed(CarA)**.
4. Resolve Proceed(CarA) with ¬Proceed(CarA) to obtain □.

Conclusion

Proceed(CarA) is proved true.

(c) Analysis for Two Emergency Vehicles

The current KB can infer GreenSignal for multiple emergency vehicles independently but cannot resolve conflicts when two emergency vehicles arrive simultaneously.

Additional Rule

∀x∀y [EmergencyType(x) ∧ EmergencyType(y) ∧ Priority(x) > Priority(y) → GreenSignal(x) ∧ Wait(y)]

Additional Facts

- Priority(Ambulance1)=2
- Priority(FireTruck1)=1

Justification

Priority rules prevent contradictory green signals and ensure safe emergency routing.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

(a) CNF

Conversion P1

SoilDry → IrrigationNeeded

Eliminate implication:

¬SoilDry ∨ IrrigationNeeded

P2

IrrigationNeeded ∧ CropWheat → ApplyDripMethod

Eliminate implication:

¬(IrrigationNeeded ∧ CropWheat) ∨ ApplyDripMethod

Apply De Morgan's law:

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

P3

¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Eliminate implication:

¬(¬ApplyDripMethod ∧ CropWheat) ∨ CropAtRisk

Apply De Morgan's law:

ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

Facts

- SoilDry
- CropWheat

Final CNF Set

1. ¬SoilDry ∨ IrrigationNeeded
2. ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
3. ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
4. SoilDry
5. CropWheat

(b) Resolution Refutation for ApplyDripMethod

Negated goal:

¬ApplyDripMethod

Resolution

1. Resolve (¬SoilDry ∨ IrrigationNeeded) with SoilDry → **IrrigationNeeded**
2. Resolve (¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod) with IrrigationNeeded → **¬CropWheat** ∨ **ApplyDripMethod**
3. Resolve with CropWheat → **ApplyDripMethod**
4. Resolve with ¬ApplyDripMethod → □

Resolution Tree

SoilDry

|

¬SoilDry ∨ IrrigationNeeded

|

IrrigationNeeded

|

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

|

¬CropWheat ∨ ApplyDripMethod

|

CropWheat

|

ApplyDripMethod

|

¬ApplyDripMethod

|

□

Conclusion

ApplyDripMethod is proved.

(c) Removing SoilDry

Remaining facts:

- CropWheat

Without SoilDry, IrrigationNeeded cannot be derived. Therefore ApplyDripMethod cannot be derived. Since P3 requires ¬ApplyDripMethod as a fact, CropAtRisk also cannot be concluded.

Conclusion

After removing SoilDry, neither ApplyDripMethod nor CropAtRisk follows from the KB.

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

(a) Belief State and Modus

Ponens Percepts

- Stench(1,2)
- Breeze(1,1)
- Glitter(2,2)

Rules

1. Stench(1,2) → Wumpus adjacent to (1,2)
2. Breeze(1,1) → Pit adjacent to (1,1)
3. Glitter(x,y) → Gold(x,y)

Derived Facts

- Wumpus adjacent to (1,2)
- Pit adjacent to (1,1)
- Gold at (2,2)

(b) Forward

Chaining From

Stench(1,2)

Adjacent cells: (1,1), (1,3), (2,2)

Possible Wumpus locations:

- W(1,1)
- W(1,3)
- W(2,2)

From Breeze(1,1)

Adjacent cells: (1,2), (2,1)

Possible Pit locations:

- P(1,2)
- P(2,1)

Final Possible Locations

Wumpus: {(1,1), (1,3), (2,2)}

Pit: {(1,2), (2,1)}

(c) Safety of Path [1,1] → [1,2] → [2,2]

- Cell (1,2) is a possible pit.
- Cell (2,2) is a possible Wumpus location.
- Gold is present at (2,2), but danger is not ruled out.

Conclusion

The path cannot be guaranteed safe. The agent should not move to (1,2) or (2,2) without additional percepts or reasoning.